Final Design Document Template

SDG Dashboard



CREATING MEANINGFUL EXPERIENCES

Academy/Department

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In this template you are provided with headings which align to the DataLab tasks. The template roughly covers the phases of the CRISP-DM lifecycle, except "*Modelling*" has been replaced with "*Exploratory Data Analysis & Data Visualization*" and "*Deployment*" has been replaced with "*Publishing & End-User Documentation*" as you will not be carrying out modelling or deployment during this project.
Read the suggestions under each task carefully, which you can use to help write your report. Note: **Do not just simply answer the questions, they are there to guide your writing in each section.** You should elaborate in detail for each question at a minimum.

Task 2.1 Business Understanding

In this section, you should write an introduction to your project.

- Identify the specific SDG indicator you will focus on and explain the importance of this indicator to the broader SDG goals.
- This project analyzes **SDG Indicator 4.5.1 - Parity indices** in education, specifically focusing on gender parity in tertiary education attainment across four European countries. SDG 4.5.1 measures equality of access to education across different demographic groups (gender, location, wealth, disability) and is critical to achieving SDG 4's broader goal of ensuring inclusive and equitable quality education for all by 2030.
- Gender parity in education serves as a fundamental indicator of social equity. When men and women have equal access to higher education, societies benefit from diverse talent pools, stronger economic growth, and reduced gender-based discrimination. However, recent global trends show concerning shifts: the UN reports that gender parity in tertiary education has declined from 94.8 to 91.6, indicating men are increasingly falling behind women in higher education attainment.
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- Clearly state what you aim to achieve with the dashboard, including a problem statement and your research question.
- **Problem Statement**
- While gender parity metrics provide valuable insights into educational equity, understanding the relationship between educational attainment and quality of life outcomes across different national contexts remains underexplored. European countries vary significantly in their educational systems, economic structures, and social welfare provisions, potentially moderating how education translates into life satisfaction and economic wellbeing.
- **Research Question**
- **"How does gender parity in tertiary education (SDG 4.5.1) compare across Germany, Sweden, Serbia, and Spain from 2018-2023, and how do education levels associate with quality of life outcomes?"**
- This question examines two interconnected dimensions:
- **Gender parity**: Whether men and women have achieved equal representation in tertiary education across different European regions
- **Education-outcome associations**: How educational attainment correlates with income and life satisfaction, and whether these relationships vary by national context
-
- What insights do you hope to gain?

The dashboard aims to reveal:

- Gender representation patterns in tertiary education across Western, Northern, Eastern, and Southern Europe

- Whether education's "returns" (income, life satisfaction) differ significantly between countries
- Temporal changes in educational equity and outcomes from 2018-2023
- The extent to which country-level factors (welfare systems, economic development) moderate education's impact on wellbeing
-
- Who are your stakeholders?
- Educational policymakers in European governments who design policies to promote educational equity and evaluate returns on educational investments
- EU education authorities monitoring progress toward SDG 4 targets across member states
- University administrators developing gender-inclusive recruitment and retention strategies

Task 2.2 Data Understanding

In this section, you should write about the data for your dashboard. This should include information such as:

- Where did you got your data (include links in the references)
- The dashboard utilizes data from the **European Social Survey (ESS)**, a cross-national survey measuring attitudes, beliefs, and behavior patterns across Europe. Three ESS rounds were combined:
- **ESS Round 9 (2018)**: https://ess.sikt.no/en/study/172ac431-2a06-41df-9dab-c1fd8f3877e7
- **ESS Round 10 (2020)**: https://ess.sikt.no/en/study/c00ba7e6-1077-4da8-93f0-a0d8b0c0fb67
- **ESS Round 11 (2023)**: https://ess.sikt.no/en/study/a708eba4-7e9f-4e8e-9a52-adaef1d649e3
- Supplementary reference data from the **SDG Indicators Database** provided official parity index trends for broader context.
- Sustainable development goals Indicator: 4.5.1 - Parity indices - Statistics Denmark
- Detail each data set you have used.
    - What format are they e.g. CSV, Excel?
    - CSV files
    - What are the variables? What data types are they?
    - Country-Text, Survey round/year-Integer, Age of respondent-Integer Gender-Integer, Education level (ISCED) - Integer, Life satisfaction-Integer, Happiness level-Integer, Health status-Integer, Income decile-Integer, Feeling about income-Integer, Employment status-Integer, Trust in people-Integer, Satisfaction with economy-Integer, Satisfaction with government-Integer, State of education system-Integer.
    - Are there problems with the quality of the data?
    - **Missing value codes**: The ESS uses standardized codes for different types of non-response:
    - **55** = Not applicable
    - **66, 6666** = Not available
    - **77, 7777** = Refusal
    - **88, 8888** = Don't know
    - **99, 9999** = No answer
    - These codes were present across most variables and required removal during data preparation.
    -
    - Are there initial insights you gained?
- **Data structure complexity**: Each ESS round contained 618 variables covering diverse topics (political attitudes, social values, demographic characteristics). Filtering to relevant educational equity variables was essential.

- **ISCED framework**: The eisced variable uses the International Standard Classification of Education, enabling cross-national comparisons despite different national education systems. This framework maps education levels from 1 (Primary or below) to 7 (Master's degree).
- **Time point selection**: The three rounds (2018, 2020, 2023) provide insight into trends, though 2020 data may be affected by COVID-19 disruptions. The five-year span is sufficient to detect meaningful changes in educational attainment patterns.
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Task 2.3 Data Preparation

In this section, you should write about how you prepared your data for the dashboard. Document the steps you took to clean, transform, and prepare the data for analysis. This may include removing duplicates, handling missing values, creating new variables and / or merging data sources that will be useful for the dashboard.

Step 1: Variable Selection and Filtering

Challenge: Each ESS round contained 618 variables, most irrelevant to educational equity analysis.

Solution:

- Loaded each ESS round separately into Power Query
- Selected only the 16 relevant variables listed in Section 2.2
- Removed all other columns to improve data model performance

Step 2: Missing Value Treatment

Challenge: ESS uses numeric codes (55, 66, 77, 88, 99) to represent different types of non-response, which would distort statistical calculations if treated as actual values.

Solution:

- Created conditional columns to replace all missing value codes with null values
- Applied transformations systematically across all relevant variables:

Step 3: Country Code Transformation

Challenge: ESS stores countries as two-letter ISO codes (DE, SE, RS, ES), which are not user-friendly for dashboard audiences.

Solution:

- Replaced country codes with full country names using conditional column:
    - DE → Germany
    - SE → Sweden
    - RS → Serbia
    - ES → Spain
- Filtered dataset to include only these four countries (originally 29+ countries were present in ESS data)

Step 4: Categorical Variable Labeling

Challenge: Most ESS variables use numeric coding (e.g., gender: 1=Male, 2=Female), making visualizations difficult to interpret without extensive documentation.

Solution: Created new labeled columns while preserving original numeric values for calculations:

Gender:

- 1 → "Male"
- 2 → "Female"

Education Level:- simplified ISCED categories:

- 1 → "Primary or Below"
- 2 → "Lower Secondary"
- 3 → "Upper Secondary General"
- 4 → "Upper Secondary Vocational"
- 5 → "Post-Secondary Non-Tertiary"
- 6 → "Bachelor"

- 7 → "Master"

Health Status:

- 1 → "Very Good"
- 2 → "Good"
- 3 → "Fair"
- 4 → "Bad"
- 5 → "Very Bad"

Income Feeling:

- 1 → "Living Comfortably"
- 2 → "Coping"
- 3 → "Difficult"
- 4 → "Very Difficult"

Employment Status: - simplified from 9 categories to most common:

- 1 → "Paid Work"
- 2 → "In Education"
- 3 → "Unemployed, Actively Looking"
- 4 → "Unemployed, Not Looking"
- 5 → "Permanently Sick/Disabled"
- 6 → "Retired"
- 7 → "Community/Military Service"
- 8 → "Housework/Caring"
- 9 → "Other"
- **Step 5: Calculated Columns**

Age Groups: Binned continuous age variable into meaningful life stages:

- 18-30 → "18-30"
- 31-45 → "31-45"
- 46-60 → "46-60"
- 60+ → "60+"

Step 6: Data Filtering

- **Null removal**: Applied filters to remove rows with null values in critical variables:
- Country (0 nulls - required field)

Step 7: Appending ESS Rounds

- **Challenge**: Need to analyze trends across 2018, 2020, and 2023, but data exists in three separate files.
- **Solution**:
- Ensured consistent column names and data types across all three ESS rounds
- Used Power Query "Append Queries" to stack ESS9, ESS10, and ESS11 in one big table.
-

Task 2.4 Exploratory Data Analysis & Data Visualization

In this section, you should write about your data model, any calculations you did and the visualizations you created and the usability of your dashboard. This should include information such as:

- Any DAX that you created
- Total Respondents = COUNTROWS('ESS Table')
- Avg Life Satisfaction = AVERAGE('ESS Table'[Life Satisfaction])
- Avg Happiness = AVERAGE('ESS Table'[Happiness])
-

- Avg Income Decile = AVERAGE('ESS Table'[Income Decile])
- % Tertiary Education =
- DIVIDE(
- COUNTROWS(
- FILTER('ESS Table',
- [Education Level (ISCED)] IN {"Bachelor", "Master"}
-)
-),
- [Total Respondents],
- 0
-)
-
- Gender Parity Index =
- DIVIDE(
- CALCULATE([% Tertiary Education], [Gender] = "Female"),
- CALCULATE([% Tertiary Education], [Gender] = "Male"),
- 0
-)
- Income Gap =
- CALCULATE(
- [Avg Income Decile],
- [Education Level] IN {"Bachelor", "Master"}
-) -
- CALCULATE(
- [Avg Income Decile],
- [Education Level] IN {"Primary or Below", "Lower Secondary"}
-)
-
- Life Satisfaction Gap =
- CALCULATE(
- [Avg Life Satisfaction],
- [Education Level] IN {"Bachelor", "Master"}
-) -
- CALCULATE(
- [Avg Life Satisfaction],
- [Education Level] IN {"Primary or Below", "Lower Secondary"}
-)
- Health Status Mode =
- VAR HealthCounts =
- ADDCOLUMNS(
- VALUES('ESS Table'[Health]),
- "Count", CALCULATE(COUNTROWS('ESS Table'))
-)
- VAR MaxCount = MAXX(HealthCounts, [Count])
- RETURN
- MAXX(
- FILTER(HealthCounts, [Count] = MaxCount),
- 'ESS Table'[Health]
-)
- Descriptive statistics
- **Total respondents**:
- **Education distribution across countries**

- **Tertiary education rates** (Bachelor + Master combined):
- **Life satisfaction** (0-10 scale):
- **Income distribution** (mean decile)
-
- Correlation calculations
- There are no correlation calculations but there are gap calculations:
- **Income Gap by Country** (difference in mean Income Decile between tertiary and primary/lower secondary education):
- **Life Satisfaction Gap by Country** (difference in mean life satisfaction between tertiary and primary/lower secondary education):
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- Visualisations- what ones you have chosen & why? What do the visualisations tell you and how do they relate to your research question? Include some images.
- **KPI Cards** (5 cards at top of page): Display key summary statistics that update based on slicer selection:
- **Total Respondents**: Sample size for selected countries
- **Average Happiness**: Mean happiness score (0-10)
- **Health Status**: Most common health category (mode)
- **Average Income Decile**: Mean income position (1-10)
- **Income Feeling**: Most common income feeling (mode)
- These KPIs provide quick context about the tertiary-educated population's characteristics and help users understand the filtered dataset size and general wellbeing levels.
- **Donut Chart: % Tertiary Education by Gender**
- Shows gender split among tertiary-educated respondents
- Key finding: Near-perfect 50/50 gender parity across all four countries (typically 51-52% female, 48-49% male)
- Insight: Gender parity in tertiary educational attainment has been achieved, confirming SDG 4.5.1 target success
- **Clustered Column Charts** (3 charts showing temporal trends): All use consistent yellow (female) and blue (male) color coding with years (2018, 2020, 2023) on x-axis:
- **Happiness by Year and Gender**
- **Finding**: No significant gender gaps in happiness among tertiary-educated individuals
- **Income Decile by Year and Gender**
- **Finding**: Small income gaps persist (men 0.3-0.5 deciles higher) even at equal education level
- Gaps remained stable across time period, particularly visible in Serbia and Germany
- **Insight**: Educational parity doesn't automatically eliminate gender pay gaps
- **Life Satisfaction by Year and Gender**
- **Finding**: Minimal gender differences, similar to happiness patterns
- Both genders report comparable life satisfaction across all years
- Clustered columns chosen over line charts due to only 3 time points
- **100% Stacked Bar Chart**: Tertiary Respondents by Income Feeling and Gender
- Two horizontal bars (Male and Female) divided into four income feeling categories
- **Finding**: Women slightly more likely to report income difficulties ("Difficult" or "Very Difficult") even at equal education levels
- **Insight**: Subjective economic security differs by gender despite equal educational credentials
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- **Page 2: Education & Income Relationships**
- **Purpose**: Analyze how education level associates with economic outcomes across countries.
- **Line Chart**: Income Decile by Education Level and Country
- Four colored lines (one per country) showing education-income trajectories across all 7 education levels
- **Key findings**:

- **All countries show upward trend** - education consistently associates with higher income
- **Insight**: Education's economic "value" varies dramatically by country - welfare systems and labor markets moderate returns
- **Bar Charts** (showing income gaps):
- **Income Gap by Country**
- Horizontal bars comparing gap size across four countries
- **Finding**: 60% difference between Serbia and Sweden in education-income association strength
- **Income Gap by Year**
- Shows temporal trends in gaps from 2018-2023
- **Finding**: Gaps peaked in 2020
- **Gauge Chart**: Overall Income Gap
- Displays average gap across all countries
- Provides single summary statistic for quick reference
- **Clustered Bar Chart**: Education Distribution by Country
- Shows proportion of population at each education level per country
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- **Page 3: Education & Life Satisfaction**
- **Purpose:** Examine how education associates with subjective wellbeing and other quality of life outcomes, distinct from economic measures.
- **Line Chart:** Life Satisfaction by Education Level and Country
- **Dramatic country differences:**
- **Critical insight:** Education-satisfaction relationship varies even more dramatically than education-income relationship, demonstrating that country context (welfare systems, economic development) fundamentally shapes whether education translates to wellbeing
- **Matrix Tables (3 tables showing additional quality of life indicators):**
- **Health by Education Level**
- Shows modal health status category for each education level
- **Insight:** Education associates with better health self-assessment, likely due to health literacy, access to healthcare, and healthier lifestyles
- **Happiness by Education Level**
- Displays average happiness scores (0-10) for each education level
- **Insight:** Happiness shows more compressed distribution than life satisfaction - education matters but less dramatically than for life evaluation
- **Economy Satisfaction by Education Level**
- Shows satisfaction with national economy (0-10) by education level
- **Pattern:** Clear gradient with higher education associated with more positive economic perceptions
- **Insight:** Better-educated individuals view economic conditions more favorably, possibly due to personal economic security or different information sources/political views
- Bar Chart: Life Satisfaction Gap by Country
- **Interpretation:** Quantifies the dramatic pattern visible in line chart - education's association with life satisfaction depends entirely on national context
- UX design, what features have you included to increase the usability of your dashboard?
- **Page navigation bar** Three-button vertical navigation panel enabling quick movement between dashboard pages
- **Country Slicer** (appears on all three pages):

Task 2.5 Evaluation

In this section you will write about the success of your dashboard. This should include information such as:

- Evaluate the results of your modelling or visualization efforts
- Discuss whether the dashboard meets the original objectives set out in the business understanding phase
- How might the research impact different stakeholders or communities?
- Identify any limitations and how they might impact the dashboard's effectiveness.
- How could your dashboard be improved? For example, if you were to continue developing the dashboard into the future, what other date could be added?

Task 2.6 Publishing & End User Document

Publish your dashboard and include a link here. You must also prepare a separate end user document detailing how to use your dashboard.
SDG Dashboard - SDG 4 Dashboard Valeri Kamenov 251041 - Power BI

End User Document - https://github.com/BredaUniversityADSAI/2025-26a-fai1-adsai-ValeriKamenov251041/blob/main/Deliverables/Project/End%20User%20Document.pdf

References

List the references used in the report here.



Games



Leisure & Events



Tourism



Media



Data Science & AI



Hotel



Logistics



Built Environment



Facility

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